REG.NO.:

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

CONTINUOUS ASSESSMENT TEST - I

WINTER SEMESTER 2024-2025

SLOT: D1

Programme Name & Branch	**: MCA**
Course Code and Course Name	**: PMCA602L & Python Programming**
Faculty Name(s)	**: Dr. Kumaresan P, Dr. Selva Rani B**
Class Number(s)	**: VL2024250503360, VL202425053365**
Date of Examination	**: 30 January 2025**
Exam Duration	**: 90 minutes** **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- You are expected to provide the **sample input/output** for all the python codes
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes
  CO1 – Develop solutions using the basic programming constructs and data structures in Python

Q. No	Question	M	CO	BL
1.	a) Write a Python code to check whether the given number is a Factorial Number or not.	5	1	3
	b) Display the following pattern using Python: # # # # # # # # # # # # #	5		
2.	Given S1 = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} S2 = {0, -6, 7, -3, 2, 1, -10} Predict the output of the following SET operations: a) S1.difference(S2) b) S1.symmetric_difference(S2) c) S1.intersection(S2) d) S1.intersection_update(S2) e) S1.difference_update(S2)	10	1	2
3.	a) Write a Python Code to replace all consonants in the given string with '$'	5	1	2
	b) Read a string with combination of characters and digits. Write a Python code to find the sum of the digits in the given string.	5		

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4.	Read two integers 'x' and 'y' (x < y). Write a Python code to generate two separate lists: a) 'Fib' with the numbers in Fibonacci series between 'x' and 'y' (5 Marks) b) 'Perf' with the Perfect numbers between 'x' and 'y' (5 Marks)	10	1	3

5.	Create the following nested dictionary:
	
	LIBRARY = {
	"B1": {"TITLE": "Clean Code: A Handbook of Agile Software Craftsmanship", "AUTHOR": "Robert C. Martin", "PRICE": 40.99, "PUBLISHER": "Prentice Hall"},
	"B2": {"TITLE": "Design Patterns: Elements of Reusable Object-Oriented Software", "AUTHOR": "Erich, Richard", "PRICE": 47.99, "PUBLISHER": "Addison-Wesley"},
	"B3": {"TITLE": "The Pragmatic Programmer: Your Journey to Mastery", "AUTHOR": "A Hunt, D Thomas", "PRICE": 42.99, "PUBLISHER": "O'Reilly Media"},
	"B4": {"TITLE": "Artificial Intelligence: A Modern Approach", "AUTHOR": "Stuart Russell, Peter Norvig", "PRICE": 89.99, "PUBLISHER": "Pearson"}
	}
	a) Read the name of the author and display the details of the book if it is available in the LIBRARY. [3 Marks]
	b) If not available, read the required details and add the new entry into the dictionary. [4 Marks]
	c) Read the name of the publisher and display the total number of books published by them. [3 Marks]
		10	1	3

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